

Chinese-Jev: Bringing System One Model to Chinese-Language Tasks

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Abstract

System One models such as Jev offer an efficient alternative to generative language models for tasks that require decisions rather than open-ended responses. However, existing Jev models exhibit limited Chinese-language decision accuracy, restricting their utility in both general and specialized settings. In this paper, we introduce CHINESE-JEV, a System One model that addresses this gap through a unified data processing and training pipeline. Our data processing protocol converts heterogeneous Chinese-language annotations into probability targets over candidate options, enabling a shared training formulation across domains and question formats. To enable efficient inference, Chinese-Jev adopts a lightweight encoder-only backbone for text encoding and learns to score candidate answers through decision-oriented training. To address the misalignment between the pre-training distribution and downstream Chinese-language scenarios, we first train the model on a general-purpose corpus of 10 million examples, then fine-tune it separately for the medical, legal, and financial domains. To evaluate decision accuracy and calibration in both general and domain-specific Chinese-language settings, we introduce Chinese-Jev Bench (CJ-Bench). After first-stage pre-training, Chinese-Jev exceeds the accuracy of the closed-source Jev model by 1.24% on general-domain tasks while achieving a $20.3\times$ speedup. Subsequent domain-specific fine-tuning yields a 4.0% accuracy improvement over Jev in medicine and retains 92% of Jev’s average accuracy across specialized domains, with a $17\times$ speedup and an average latency of only 15 ms per example. We further demonstrate on-device deployment of an INT8-quantized model on mobile devices, achieving an inference latency of approximately 1 second per decision. We will release the models, training data, benchmark, and data construction code. The project is available at <https://gulu captain.github.io/Chinese-Jev/>.

1 Introduction

An application does not always need a language model to write a response. It may need to route a request to a service (Larson et al., 2019; Casanueva et al., 2020), judge the relevance of a passage (Nogueira and Cho, 2019; Khattab and Zaharia, 2020), or select an answer from a set of alternatives (Lai et al., 2017; Sun et al., 2020). These tasks require language understanding, but their outputs are bounded. Large language models (LLMs) offer a flexible way to specify such tasks through instructions and examples (Wei et al., 2022; Sanh et al., 2022; Longpre et al., 2023). Repeatedly invoking a large model, however, can be costly when an application needs only a label or a score. The practical goal is to retain flexibility across tasks while reducing the cost of each decision.

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Jev is a System One model designed for three forms of decision making: *choice* selects among candidate alternatives, *noul* determines the validity of a proposition, and *score* assigns an ordered rating (Almeida, 2026). Its structured outputs and associated probabilities can be used directly by software. Alongside the hosted Jev API, open implementations explore compact encoders and language-model-based decision systems (Kotoba Labs, 2026; NandhaKishorM, 2026; Cai, 2026; jaredpalmer, 2026; Lee, 2026). Chinese-oriented releases include a model trained primarily on football-domain data (xuhaodev, 2026) and a bilingual encoder for local agent decisions.

Building on these efforts, we introduce CHINESE-JEV, a compact System One model for Chinese-language decision-making across general tasks and the medical, legal, and financial domains. Central to this work is a unified supervision formulation that accommodates heterogeneous decision tasks. Existing Chinese benchmarks and instruction collections provide diverse source data (Xu et al., 2020; Bai et al., 2025), but categorical labels, proposition judgments, and ordered ratings encode different forms of supervision. Our data processing protocol converts these annotations into probability distributions over candidate options while preserving label semantics and rating order. It also incorporates soft supervision when multiple ratings are available. Applying this protocol yields a general corpus of 10 million decisions spanning eight task categories, together with dedicated corpora for medicine, law, and finance. Figure 1 summarizes the composition of the general corpus. To reduce leakage between training and evaluation, decisions derived from the same source material are assigned to the same data partition. This formulation supports option selection, proposition judgment, and ordinal rating through a shared candidate-scoring interface in a single forward pass.



Figure 1: General training corpus composition across eight task categories, ordered alphabetically. Stacked bars indicate category proportions, with colors denoting choice, noul, and score.

Moreover, to address the mismatch between multilingual pre-training and downstream Chinese-language tasks while maintaining efficient inference, we adopt a two-stage training pipeline built on a lightweight encoder-only backbone (Marone et al., 2025). We first train CHINESE-JEV GENERAL on a corpus of 10 million Chinese-language decisions spanning eight task categories, then independently fine-tune it on medical, legal, and financial corpora to obtain three domain specialists. All models retain the same architecture and decision interface, enabling domain specialization without increasing model size or changing how applications specify decisions.

We further introduce Chinese-Jev Bench (CJ-Bench), comprising 307,900 held-out decisions across general, medical, legal, and financial tasks, to evaluate decision accuracy, calibration, and inference latency under a common protocol. After first-stage pre-training, Chinese-Jev achieves 69.20% accuracy on the general subset, exceeding the closed-source Jev model by 1.24% in relative accuracy and reducing expected calibration error from 11.45% to 3.78%. Its measured latency is $20.3\times$ lower than that of the hosted Jev API. Following domain-specific fine-tuning, Chinese-Jev improves medical accuracy over Jev by 4.0% and achieves 92% of Jev’s average accuracy across the three specialized

domains, with an average latency of 15 ms per decision and a $17\times$ reduction in measured latency. Finally, an INT8-quantized deployment enables local inference in mobile devices at approximately 1 second per decision, demonstrating the feasibility of on-device Chinese-language decision-making.

Our main contributions are:

- **Chinese-Jev: a System One model for Chinese-language decision-making.** We develop Chinese-Jev through general Chinese-language pre-training followed by independent fine-tuning in medicine, law, and finance. The resulting general and specialist models share a unified decision interface that supports option selection, proposition judgment, and ordinal rating in a single forward pass.
- **A unified pipeline for Chinese-Jev training data construction.** We design a reusable pipeline that converts heterogeneous Chinese QA annotations into probability targets over candidate options for Chinese-Jev training. Using this pipeline, we construct a pre-training corpus of 10 million examples spanning eight task categories, together with downstream fine-tuning corpora.
- **CJ-Bench, empirical evaluation, and on-device deployment.** We introduce CJ-Bench, comprising 307,900 held-out decisions, to evaluate accuracy, calibration, and latency across general and specialized tasks. Chinese-Jev surpasses the closed-source Jev model in general-task accuracy and achieves 92% of its average accuracy across specialized domains, with measured speedups of $20.3\times$ and $17\times$, respectively, relative to the hosted Jev API. An INT8 browser deployment further enables local smartphone inference at approximately 1 second per decision.

2 Related Work

2.1 Generalist Text Classification

Generalist text classification uses natural-language label descriptions to predict across tasks and label sets (Yin et al., 2019; Laurer et al., 2023; Stepanov et al., 2025). Input-label embedding methods learn compatibility between texts and labels to transfer to unseen categories (Pappas and Henderson, 2019). Natural language inference (NLI) classifiers treat the input as a premise and each candidate label as a hypothesis (Yin et al., 2019). Universal NLI classifiers combine entailment data with classification datasets reformulated as premise-hypothesis pairs (Laurer et al., 2023). GLiClass jointly processes the input and all candidate labels in one encoder pass, allowing both text-label and label-label interactions (Stepanov et al., 2025). Chinese-Jev also encodes the input and candidate answers jointly, with separate readouts for choices, proposition judgments, and ordered ratings.

2.2 Jev and Typed Decision Models

Jev exposes choice, noul, and score decisions together with their associated probabilities (Almeida, 2026). Kotoba Labs’ Open-Jev and Laya learn candidate-scoring functions over bidirectional encoder representations, with Laya’s multilingual variant using mmBERT (Kotoba Labs, 2026; NandhaKishorM, 2026). Among language-model approaches, Zefan Cai’s Open-Jev adapts a Qwen backbone with decision training (Cai, 2026), Kev adds learned decision heads (jaredpalmer, 2026), and Nimble fine-tunes predictions over allowed answer tokens (Bespoke Labs and Sathiamoorthy, 2026). SemIF reads probabilities from option-token logits (Lee, 2026), while AnyJev supports debiasing, post-hoc calibration, and lightweight fitted heads (Zhang et al., 2026). Chinese-oriented releases include Qwen3-1.7B-Jev, which uses a language-model backbone and a learned decision head for football questions (xuhaodev, 2026), and MacJev, a bilingual encoder for tool routing and task-state checks (chaoliangUNSW, 2026). Chinese-Jev trains an encoder on general Chinese decisions and then adapts it separately to medicine, law, and finance.

2.3 Multitask Supervision and Chinese Resources

Multitask instruction tuning combines tasks through natural-language descriptions and shared training formats, enabling transfer beyond the tasks used for fine-tuning (Wei et al., 2022; Sanh et al., 2022; Wang et al., 2022). The Flan Collection examines task balancing, prompt diversity, and task reformulation, and shows the value of instruction-tuned checkpoints for subsequent task-specific adaptation (Longpre et al., 2023). For Chinese, CLUE covers general language understanding (Xu et al., 2020), while C-Eval and CMMLU assess knowledge and reasoning across disciplines (Huang

et al., 2023; Li et al., 2024b). COIG-CQIA provides instruction-following data from real-world sources (Bai et al., 2025). Specialized resources address biomedical language understanding, legal case retrieval, and financial knowledge and applications (Zhang et al., 2022; Li et al., 2024a; Zhu et al., 2024). Chinese-Jev converts selected source annotations into candidate-probability targets, preserving categorical labels, the order of grades, and distributions of human ratings.

3 Methods

We develop CHINESE-JEV to adapt compact System One decision models to broad Chinese-language tasks while retaining a unified interface across general and specialized domains. A unified data construction pipeline (Section 3.1) maps diverse source annotations to probability distributions over candidate options while preserving their decision semantics. A general-to-domain training pipeline (Section 3.2) first adapts the model to broad Chinese supervision and then independently specializes it for medicine, law, and finance. Together, these components provide a common training and inference interface for choice, noul, and score decisions.

3.1 Data Construction Pipeline

Figure 2 demonstrates our data construction pipeline, which consists of source discovery, annotation conversion, deduplication, and mixture construction. We first identify Chinese datasets whose annotations can be expressed as candidate-based decisions. Each source annotation is then converted into a common representation consisting of a context, an instruction, a decision type, a candidate set, and a target probability distribution. After removing duplicate and conflicting supervision, we construct the mixtures according to task coverage, available supervision, and decision-type budgets.



Figure 2: Data construction pipeline comprising source discovery, annotation conversion, deduplication, and mixture selection. Examples illustrate mapping a “DuReader Yes label” to a noul stance target and deduplicating a C3 question across candidate permutations. The general training mixture spans eight task categories with approximately balanced budgets across the three decision types.

Source discovery and selection. We identify candidate datasets from published papers, author-maintained repositories, and Hugging Face using combinations of Chinese-language, task-specific, and domain-specific keywords. Aggregated collections are traced to their original sources, and we retain datasets with Chinese content, clear usage terms, and annotations that can be expressed as candidate-based decisions or ordered scores. Representative sources include C3 for reading comprehension Sun et al. (2020), T2Ranking for relevance (Xie et al., 2023), ASAP for review judgments (Bu et al., 2021), and USTS for similarity (Wang et al., 2023), alongside domain sources such as CMB (Wang et al., 2024), LeCaRDv2 (Li et al., 2024a), and FinRE (Li et al., 2019). We further construct 10,700 rule-based decisions for condition checking, counting, and grading, with construction details provided in Appendix A.3.

Conversion to decision supervision. Each decision consists of a context x , an instruction u , a type t , a candidate list $C = (c_1, \dots, c_K)$, and a target distribution y over C . Figure 3 illustrates four routes from source annotations to this format.

(a) Single-answer question

C3

“材料：小李，快，小王的电话。
问题：谁接电话？
A：小李 B：小李的朋友 C：小王
Source answer：小李”

choice $y_A = 1, y_B = y_C = 0$

(b) Yes/No annotation

DuReader

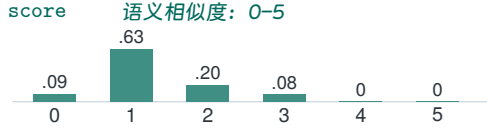
“问题：衬衫能改短吗
回答：可以
Source label：Yes”

noul 回答是否表达肯定立场？
 $y_{\text{true}} = 1, y_{\text{false}} = 0$

(c) Similarity ratings

USTS

“句子一：这是世界上所有监狱中最好的，最大的
志愿者队伍。
句子二：另外一件令我震撼的事，便是那庞大的志
愿者队伍。
19 ratings; mean ≈ 1.278947 ”

**(d) Multiple-answer question**

CMB

“心房颤动的临床特点包括
A：心律绝对不规则
B：心音强弱不等
C：脉搏短绌
D：仅见于器质性心脏病
E：心排血量下降 25%
Source answer set includes C”

noul 选项 C 是否应被选中？
 $y_{\text{true}} = 1, y_{\text{false}} = 0$

Figure 3: Conversion of source annotations into decision targets: (a) single-answer questions to choice; (b) Yes/No annotations to noul; (c) ratings to score; and (d) multiple-answer questions to option-wise noul. DuReader labels answer stance. In (c), bars show the soft target obtained from 19 human ratings, rounded to two decimals. Examples are shortened for display.

Single-answer questions and category labels $\rightarrow$ choice. For single-answer questions, we retain the original question and alternatives and assign probability one to the annotated answer. For categorical tasks, the candidate set comprises the source dataset’s labels, with all target probability mass assigned to the annotated class.

True/false and proposition annotations $\rightarrow$ noul. We express the annotated judgment as an explicit proposition and map its binary label $z \in \{0, 1\}$ to $\mathbf{y} = (1 - z, z)$, ordered as false and true. For DuReader, the proposition asks whether a supplied answer expresses an affirmative stance: a Yes label maps to true without asserting the answer’s factual correctness. Examples labeled Depends remain in the three-way choice task and are excluded from this noul conversion.

Ratings $\rightarrow$ score. For discrete ratings, we retain the original ordered scale and assign all target mass to the annotated level; relevance and quality grades follow the same rule. For continuous ratings, as in USTS, we distribute probability mass between adjacent integer levels for each rating and then average across raters. The soft target preserves the mean rating and inter-rater variation.

Multiple-answer questions $\rightarrow$ option-wise noul. We decompose each multiple-answer question into option-membership propositions. For a source question with K options, let $A \subseteq \{1, \dots, K\}$ be the correct option set and $z_j = \mathbf{1}[j \in A]$ indicate whether option j belongs to it. The target is:

$$\mathbf{y}^{(j)} = (1 - z_j, z_j), \quad (1)$$

where the entries correspond to false and true. Each decision retains the full question and all alternatives but predicts membership for a single option. Complete multi-label annotations support the same conversion.

Inference annotations support relation selection or entailment judgments, while aspect annotations support mention detection and sentiment decisions. Synthetic rule tasks use the same formats, with targets computed from the underlying rules (Appendix A.3).

Decision-level deduplication. For the general corpus, we define each decision by its task identifier r , type t , context x , instruction u , and candidates C , while excluding the target distribution so

that conflicting supervision can be detected. After Unicode and whitespace normalization N , we canonicalize the candidates as:

$$\tilde{C}_t = \begin{cases} \text{sort}(N(C)), & t = \text{choice}, \\ N(C), & t \in \{\text{noul}, \text{score}\}. \end{cases} \quad (2)$$

and compute the fingerprint:

$$h = H(r, t, N(x), N(u), \tilde{C}_t). \quad (3)$$

where H denotes a hash function. Sorting makes `choice` fingerprints invariant to permutations of independent alternatives while preserving sensitivity to their content; the order of `score` levels and the false/true semantics of `noul` remain unchanged. For matching fingerprints, we align `choice` targets by candidate text and retain a single decision when the targets agree, discarding the entire group when they conflict. Decisions that share source material, such as different questions derived from the same passage, may remain distinct but are assigned to the same data partition. Appendix A.2 describes additional cases and checks against held-out material.

Task coverage and type balance. Decision-type budgets are defined separately for each corpus. For the general corpus, we allocate approximately one third of the training decisions to each of `choice`, `noul`, and `score`; within each type, source-level budgets preserve coverage of smaller tasks and diverse data sources before the remaining capacity is filled with eligible decisions. The legal and financial corpora follow the same balanced design, whereas the medical corpus contains a larger `noul` share due to its matching and multiple-label tasks; detailed domain statistics are reported in Appendix A.4. We characterize the general corpus using eight task categories defined by prediction target and annotation semantics (Figure 1); for example, semantic matching includes similarity, equivalence, and retrieval relevance, while reading and reasoning cover contextual questions, logical reasoning, and explicit-rule decisions.

Data partitions and CJ-Bench. Each corpus has separate training, development, calibration, and test partitions, with all decisions derived from the same source material assigned to a single partition to reduce source-level leakage. Development data are used for model validation, while the calibration and test partitions remain disjoint from training and development. CJ-Bench is constructed exclusively from the held-out test partitions of the general, medical, legal, and financial corpora. Its General component covers all eight task categories with approximately equal decision-type budgets. Appendix A.1 provides the complete partition statistics and benchmark composition.

3.2 Chinese-Jev Training

CHINESE-JEV adopts a lightweight encoder-only architecture for low-latency candidate scoring across general and domain-specific tasks. Given a decision (x, u, t, C) , a bidirectional mMBERT encoder (Marone et al., 2025) jointly represents the input and candidate options, followed by a type-conditioned decision head that produces a probability distribution $\mathbf{p}$ over the candidates. We first train Chinese-Jev General on the general Chinese corpus and then independently specialize it for the medical, legal, and financial domains. The same architecture and training objective are retained throughout, preserving a consistent decision interface across all stages.

Model architecture. We serialize the decision type t , instruction u , all candidates C , and context x into a single sequence S , placing a marker at position m_i before each candidate c_i .

As shown in Figure 4, the encoder E_θ contextualizes the complete sequence:

$$H = E_\theta(S). \quad (4)$$

We then add a learned type embedding e_t at every position and apply the decision Transformer D_θ :

$$\tilde{H} = D_\theta(H + \mathbf{1}e_t^\top).$$

A shared MLP f_θ scores all candidate-marker representations in a single forward pass, with a softmax over the resulting logits yielding the decision probabilities:

$$s_i = f_\theta(\tilde{H}_{m_i}), \quad p_i = \frac{\exp(s_i)}{\sum_{j=1}^K \exp(s_j)}. \quad (5)$$

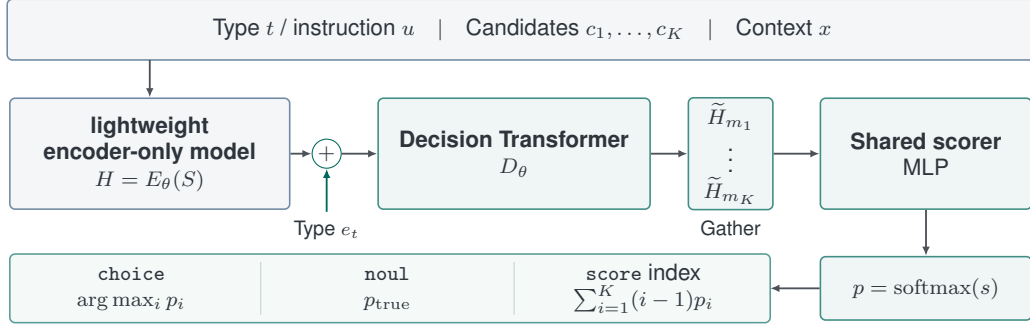


Figure 4: Chinese-Jev architecture. A lightweight encoder-only model jointly processes the input and candidates, followed by a type-conditioned decision Transformer and a shared candidate scorer. All candidate probabilities are obtained in one forward pass. The score readout is an expected zero-based level index; source-scale ratings use the associated numerical level values.

Decision readouts. For choice, we return $\arg \max_i p_i$; for noul, we return p_{true} . The score readout is the expected zero-based level index, $\sum_{i=1}^K (i-1)p_i$. On the source rating scale, the expectation is $\sum_i v_i p_i$, where v_i is the numerical value of level i .

Training objective. We train the predicted distribution $\mathbf{p}$ against a target distribution $\mathbf{y}$ over the candidates defined in Section 3.1. Following Laya’s RLCD formulation (NandhaKishorM, 2026), the objective combines supervised cross-entropy with a policy-gradient term over perturbed candidate logits. Both terms use $\mathbf{y}$. For a batch of B decisions, where decision b has K_b candidates, the cross-entropy loss is:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B} \sum_{b=1}^B \sum_{i=1}^{K_b} y_{bi} \log p_{bi}. \quad (6)$$

RLCD draws centered Gaussian perturbations of each decision’s logits, reusing the same forward pass. Its reward uses log and spherical scores to measure agreement with the target distribution. For ordered score decisions, the reward also includes the ranked probability score (RPS) to account for distance along the ordered levels. The total objective is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{RL}}. \quad (7)$$

We provide the full RLCD perturbation, reward, and detached-advantage formulation in Appendix A.5, and report the training configuration in Section 4.1.

4 Experiments

4.1 Experimental Setup

Implementation details. We initialize the mmBERT encoder and decision head from Laya Multilingual (NandhaKishorM, 2026; Convai Innovations, 2026). The backbone is mmBERT-base with 22 layers and a hidden size of 768. The head contains two decision Transformer layers and a shared candidate scorer, bringing the model to approximately 322 million parameters. Training uses full-parameter fine-tuning on eight NVIDIA H200 GPUs. We first train for one epoch on the general corpus to obtain Chinese-Jev General. Starting independently from this checkpoint, each specialist is trained for four epochs on its corresponding domain corpus. Medical specialization uses 3,000,000 decisions, while legal and financial specialization each use 24,000. The RLCD objective samples four logit perturbations per decision in both stages, shown in Appendix A.5.

Benchmark and evaluation metrics. All comparisons use CJ-Bench (Section 3.1), comprising General, Medical, Legal, and Finance components with 100,000, 200,000, 4,300, and 3,600 decisions, respectively. All models receive the same retained context and candidates and are evaluated on the same eligible examples within each component. We mainly use three metrics for evaluation: 1) Accuracy (ACC) measures agreement between the highest-probability candidate and the gold

Table 1: Results on the general subset of Chinese-Jev Benchmark. Chinese-Jev General is evaluated after first-stage pre-training. The best and second-best results are shown in **bold** and underlined, respectively; ties share the same rank. † Jev uses a hosted API, and its latency includes network round-trip time.

Model	ACC (%) ↑	ECE (%) ↓	Latency (ms) ↓
Qwen3.5-2B (Qwen Team, 2026)	32.95	43.68	280
Open-Jev (Cai, 2026)	52.33	<u>10.54</u>	60
SemIF (Lee, 2026)	44.21	27.27	<u>57</u>
Laya Multilingual (Convai Innovations, 2026)	40.22	23.04	14
Jev [†] (Almeida, 2026)	<u>68.35</u>	11.45	284
Chinese-Jev General	69.20 (+0.85)	3.78 (-6.76)	14

label. **2)** Expected calibration error (ECE) measures the gap between this candidate’s probability and empirical accuracy using 15 equal-width bins (Guo et al., 2017). ACC and ECE apply to single-label examples, including hard score targets; soft targets are excluded. We retain each checkpoint’s native probability transformations and apply no additional calibration on the target datasets. **3)** Latency is the median end-to-end time per decision, measured on one NVIDIA H200 GPU at batch size one after warm-up. Timing includes input preparation and inference through the final candidate probabilities. For the hosted Jev API, we instead measure client-observed latency for one decision per request, including network round-trip time.

Baselines. Qwen3.5-2B (Qwen Team, 2026) is the general-purpose LLM baseline, evaluated on text inputs in non-thinking mode by scoring candidate labels. Open-Jev (Cai, 2026) denotes Zefan Cai’s implementation, which adapts a Qwen backbone with decision training. SemIF (Lee, 2026) reads decision probabilities from pretrained language models. Laya Multilingual (Convai Innovations, 2026) uses a compact encoder. We access the official Jev model (Almeida, 2026; TypeSafe, 2026) through its hosted API. Each system supplies a candidate distribution for evaluation.

4.2 Experimental Results

Pre-training evaluation on general data. Table 1 demonstrates the effectiveness of first-stage Chinese-language pre-training. Starting from Laya Multilingual, Chinese-Jev improves accuracy from 40.22% to 69.20%, a gain of 28.98 percentage points, while reducing ECE from 23.04% to 3.78%. These improvements are achieved without changing the model architecture or increasing measured inference latency: both models require a median of 14 ms per decision. This comparison shows that Chinese-language training substantially improves decision accuracy and calibration without introducing additional inference cost. Chinese-Jev achieves the highest accuracy and lowest ECE among the evaluated models, while matching the lowest measured latency. It exceeds the closed-source Jev model by 0.85 percentage points in accuracy, corresponding to a 1.24% relative improvement, and reduces ECE by 67.0%, from 11.45% to 3.78%. Under the same local timing protocol, Chinese-Jev is $20\times$ faster than Qwen3.5-2B while achieving substantially higher accuracy. Its measured latency is also $20.3\times$ lower than that of the hosted Jev API, although this comparison includes network round-trip time for Jev and therefore reflects end-to-end response latency rather than a direct comparison of model inference speed.

Domain-specific evaluation. Table 2 shows that domain-specific fine-tuning substantially improves decision accuracy beyond general Chinese-language pre-training. Relative to Chinese-Jev General, the specialists improve accuracy by 47.88, 21.34, and 29.33 percentage points on Medical, Legal, and Finance, respectively, raising average accuracy from 37.97% to 70.82%. General pre-training alone provides uneven benefits across domains: on Finance, Chinese-Jev General achieves 35.39% accuracy, below the multilingual initialization’s 38.31%. These results highlight the value of supervision tailored to the target domain beyond broad Chinese-language training.

Chinese-Jev Specialist outperforms all four open baselines in accuracy across the three domains. On Medical, it achieves 87.65% accuracy, exceeding the closed-source Jev model by 3.36 percentage points, or 4.0% in relative terms. The Legal and Finance specialists reach 60.10% and 64.72%, respectively, after fine-tuning on 24,000 decisions per domain, although both remain below Jev.

Table 2: Domain-specific results on CJ-Bench. Chinese-Jev-G and Chinese-Jev-S denote the general model and the corresponding domain specialist, respectively. ACC and ECE are reported in %, and latency (Lat.) in ms. Avg. denotes the unweighted mean of domain-level metrics. The best and second-best results are shown in **bold** and underlined, respectively. † Jev’s hosted API latency includes network round-trip time.

Model	Medical			Legal			Finance			Average		
	ACC ↑	ECE ↓	Lat. ↓	ACC ↑	ECE ↓	Lat. ↓	ACC ↑	ECE ↓	Lat. ↓	ACC ↑	ECE ↓	Lat. ↓
Qwen3.5-2B (Qwen Team, 2026)	53.88	28.71	263	31.53	44.82	438	38.11	49.12	266	41.17	40.88	322.33
Open-Jev (Cai, 2026)	73.08	2.88	53	44.07	23.15	110	48.11	<u>10.19</u>	73	55.09	<u>12.07</u>	78.67
SemIF (Lee, 2026)	62.17	10.39	<u>51</u>	43.07	19.99	<u>49</u>	47.89	18.45	<u>54</u>	51.04	16.28	<u>51.33</u>
Laya (Convai Innovations, 2026)	24.55	43.05	15	34.72	31.49	15	38.31	26.18	15	32.53	33.57	15.00
Jev† (Almeida, 2026)	<u>84.29</u>	<u>5.70</u>	267	68.35	8.03	256	78.39	2.98	246	77.01	5.57	256.33
Chinese-Jev-G	39.77	17.33	15	38.76	<u>15.37</u>	15	35.39	14.43	15	37.97	15.71	15.00
Chinese-Jev-S	87.65	7.25	15	<u>60.10</u>	17.45	15	<u>64.72</u>	11.55	15	<u>70.82</u>	12.08	15.00

Across domains, the specialists achieve 92.0% of Jev’s average accuracy while retaining a median latency of 15 ms per decision in each domain. This corresponds to 17.5–29.2× speedups over Qwen3.5-2B under the same local timing protocol. Their average latency is also approximately one-seventeenth that of the hosted Jev API, whose measurements include network round-trip time. The calibration benefits of specialization are less consistent. ECE decreases from 17.33% to 7.25% on Medical and from 14.43% to 11.55% on Finance, but increases from 15.37% to 17.45% on Legal despite the accuracy gain. Across models, the Medical specialist achieves the highest accuracy, whereas Open-Jev obtains the lowest ECE. Moreover, the specialists’ average ECE remains higher than Jev’s (12.08% versus 5.57%). These results highlight the need to assess calibration separately from accuracy after domain adaptation.

4.3 Application: Interactive Decisions on Mobile Devices

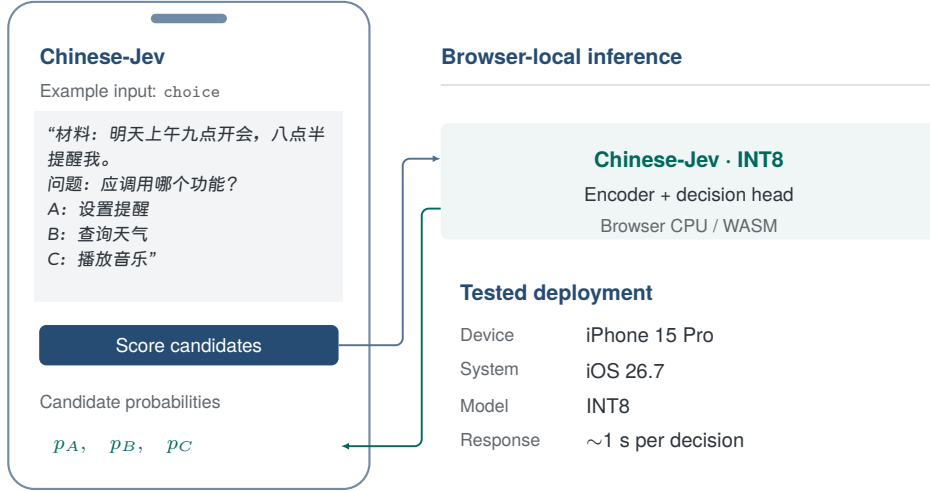


Figure 5: On-device Chinese-Jev browser prototype. A Chinese reminder request and editable candidates are scored locally by the INT8 model. The interface is schematic, with p_A , p_B , and p_C standing for returned probabilities. The tested iPhone 15 Pro configuration responds in approximately one second per decision.

Our mobile browser prototype enables users to submit Chinese questions with context and editable candidate answers, and obtain option probabilities directly on their smartphones. Figure 5 illustrates this workflow using a meeting message with three candidate actions: setting a reminder, checking the weather, and playing music. On an iPhone 15 Pro running iOS 26.7, the INT8-quantized model achieves an inference latency of approximately 1 s per decision. Both tokenization and inference execute locally in the browser, with downloaded model files cached for reuse. Once the model is loaded, subsequent queries are processed entirely on-device without transmitting input text to a remote inference service.

5 Conclusion

We introduced CHINESE-JEV, a System One model for Chinese-language decision-making across general tasks and specialized domains. Our unified data construction pipeline supports large-scale Chinese pre-training and subsequent fine-tuning in medicine, law, and finance. We also introduced CJ-Bench to evaluate decision accuracy, calibration, and inference latency. Chinese-Jev surpasses the closed-source Jev model on general tasks and in medicine, while achieving 92% of its average accuracy across specialized domains. Measured latency improves by approximately $20.3\times$ on general tasks and $17\times$ across specialized domains relative to the hosted Jev API, whose latency includes network overhead. An INT8 mobile browser deployment further demonstrates on-device inference at approximately 1 second per decision. Remaining accuracy gaps in law and finance, together with uneven calibration gains after specialization, motivate further work on domain adaptation and confidence calibration. We will release the models, training data, benchmark, and data construction code to support reproducible research.

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A Additional Data and Training Details

A.1 Dataset Composition

Table 3 reports the corpus partitions. The four benchmark partitions form Chinese-Jev Benchmark; its General component is selected from the full General test pool. Counts refer to decisions, including those derived from a shared source record. Table 4 gives the general training contributions from 20 public sources and the programmatic rule supplement.

Table 3: Corpus partitions by decision type, counted in decisions. Rows labeled Benchmark make up Chinese-Jev Benchmark. The 100,000-decision General component is a subset of the full General test pool.

Partition	choice	noul	score	Total
<i>General</i>				
Train	3,333,334	3,333,333	3,333,333	10,000,000
Development pool	179,723	215,520	357,543	752,786
Calibration pool	63,820	96,679	107,454	267,953
Full test pool	214,513	929,814	823,368	1,967,695
Benchmark	33,334	33,333	33,333	100,000
<i>Medical</i>				
Train	1,194,496	1,792,223	13,281	3,000,000
Validation	49,416	49,416	1,168	100,000
Calibration	9,846	9,847	307	20,000
Benchmark	98,543	98,544	2,913	200,000
<i>Legal</i>				
Train	8,000	8,000	8,000	24,000
Validation	797	1,021	700	2,518
Calibration	317	450	250	1,017
Benchmark	1,500	1,500	1,300	4,300
<i>Finance</i>				
Train	8,000	8,000	8,000	24,000
Validation	600	600	600	1,800
Calibration	200	200	200	600
Benchmark	1,200	1,200	1,200	3,600

Full-input limits: 1,024 tokens for General and Finance, 2,048 for Medical, and 8,192 for Legal. Medical also uses a 256-token decision-header limit.

Table 4: Source contributions to the 10 million general training decisions, combining 20 public data sources and a programmatic rule supplement. Sources are listed in descending order of contribution down each half; percentages are rounded.

Source	Decisions	Share	Source	Decisions	Share
T2Ranking (Xie et al., 2023)	2,328,371	23.28%	CMRC2019 (Cui et al., 2020)	93,605	0.94%
ASAP (Bu et al., 2021)	1,318,889	13.19%	OCNLI (Hu et al., 2020)	59,158	0.59%
Dianping (Zhang and LeCun, 2017)	1,256,330	12.56%	PAWS-X (Chinese) (Yang et al., 2019)	45,508	0.46%
JD full reviews (Zhang and LeCun, 2017)	973,902	9.74%	Poetry retrieval (PoetryMTEB Contributors, 2026)	29,796	0.30%
DMSC (utmhikari, 2017)	933,902	9.34%	LogiQA 2 (Chinese) (Liu et al., 2023b)	11,526	0.12%
MEP-3M (Liu et al., 2023a)	898,261	8.98%	C3 (Sun et al., 2020)	11,502	0.12%
ChID (Zheng et al., 2019)	635,043	6.35%	Programmatic rules	10,700	0.11%
CMNLI (Xu et al., 2020)	587,172	5.87%	MASSIVE (Chinese) (FitzGerald et al., 2023)	10,675	0.11%
Ifeng news (Zhang and LeCun, 2017)	504,284	5.04%	ChnSentiCorp (Tan, n.d.)	7,187	0.07%
CPED (text) (Chen et al., 2022)	150,465	1.50%	USTS (TED-X) (Wang et al., 2023)	3,007	0.03%
DuReader Yes/No (Baidu, 2019)	130,717	1.31%			
Total			10,000,000	100.00%	

General benchmark component. We select 100,000 decisions without replacement from the General test pool, with approximately equal budgets for `choice`, `nou1`, and `score`. Within each type, source tasks receive equal budgets where capacity permits; smaller tasks contribute all available examples, and unused capacity is redistributed. Selection within each task is stratified by candidate count, label, and input length. Each source contributes at most 15% of the subset. Table 5 groups the selected decisions using the same eight category definitions as the training corpus, listed alphabetically. All categories are covered. Their shares reflect the number of constituent source tasks and the available test data, rather than equal category quotas.

Table 5: Task composition of the General benchmark component (100,000 decisions) and the general training corpus (10 million decisions). Categories are listed alphabetically; counts include hard and soft targets, and shares use the full size of each partition.

Task category	Train (%)	Test decisions	Test (%)
Aspect mention	4.269	4,480	4.480
Aspect status and sentiment	8.560	6,481	6.481
Category and intent	14.132	9,268	9.268
Reading and reasoning	7.624	11,784	11.784
Review rating	19.438	9,987	9.987
Semantic matching	24.067	26,295	26.295
Sentiment and emotion	14.140	11,961	11.961
Textual inference and stance	7.770	19,744	19.744
Total	100.000	100,000	100.000

A.2 Deduplication and Shared Material

Equation 3 describes exact deduplication of general decisions after task-specific conversion. Unicode NFC and whitespace normalization preserve lexical content, including numbers and negation. Equality is evaluated within a task definition, so matching text under different task identities remains separate.

For choice, target comparison pairs each normalized candidate text with its probability before sorting. A changed answer letter caused solely by reordering does not create a label conflict. Table 6 summarizes the resulting decisions.

Table 6: Exact deduplication rules for general decisions. Unless a change is specified, examples share the same task, decision type, context, and instruction. Matching inputs are merged only when their aligned targets agree; conflicting groups are discarded.

Variation between examples	Treatment
Record identifier or normalized whitespace	Same fingerprint; retain one when targets agree.
choice alternatives reordered	Same fingerprint; compare targets by candidate text, not answer letter.
Alternative added, removed, or replaced	Different decision, even when the question and correct answer are unchanged.
Same normalized input, different target	Exclude all examples with that fingerprint.
Different task, question, number, or negation	Different decision; preserve the changed meaning.
Ordered score levels changed or reordered	Different decision; the scale order is retained.
Shared passage, different questions	Retain distinct decisions in the same partition.

Position-dependent alternatives require special treatment. The general converter rejects detected references such as “A and B” or “all of the above” when no task-specific conversion resolves them, and rejects repeated candidate text within a choice question. The medical integration pipeline uses exact fingerprints that preserve candidate order, rather than the order-invariant general rule.

Split assignment operates on shared source material and its known derived examples. Distinct questions from one passage, and related views of one annotation, remain in one partition. Official held-out material takes priority over training candidates. A separate near-duplicate screen checks training candidates against indexed held-out text and excludes unresolved close matches while preserving numerical and negation differences. This screen is limited to held-out text available when a candidate is checked; paraphrases across the full training corpus are not exhaustively merged.

A.3 Rule-Based Synthetic Data

We supplement public datasets with Chinese exercises generated from templates and rules defined in this project. The general training corpus retains 10,700 such decisions (0.107%): 10,526 concern table conditions and counts, and 174 concern rule-based grades. These counts refer to decisions, since one set of facts can support several questions.

Construction. For table tasks, we sample three to six candidate plans with prices, distances, and service availability, together with explicit selection conditions. The settings include accommodation, dining, and service packages. For grading tasks, we sample the submission status of required and optional materials for fictional registration or document-submission procedures. Chinese templates present the facts and all applicable rules. Answers are computed directly from these inputs and checked independently against the rendered text, without model-generated labels.

Some facts are marked as unknown. We retain a score question only when its exact count or grade is determined, and a nou1 proposition only when its truth value is determined. A choice question can include an insufficient-information alternative. Whether the available information determines a unique grade is itself a valid nou1 question.

Examples. Figure 6 shows two outputs of the generator. In (a), only the second accommodation plan satisfies all conditions, yielding three decisions from one table. In (b), the required materials have been submitted, but the submission status of the optional receipt is unknown. The grade could therefore be either 1 or 2: the choice answer is insufficient information, and the nou1 answer to whether the grade is uniquely determined is false. No exact-grade score question is emitted for this instance.

(a) Selecting accommodation

“条件：价格不超过 130 元，距离不超过 400 米，并且包含早餐。”				noul	方案乙满足全部条件？
				满足	(true)
				score	符合条件的方案数？
				1	(levels: 0, 1, 2, 3)
				choice	符合数量属于哪类？
				恰好一个方案符合	

(b) Applying a grading rule with missing information

“必要材料：登记表、身份核验记录。 可选材料：预约凭证。 等级规则：必要材料任一未提交为 0；必要材料齐全、可选材料有缺失为 1；全部提交为 2。 记录：登记表与身份核验记录已提交；预约凭证状态未知。”		noul	能唯一确定等级？
		无法唯一确定	(false)
		choice	等级属于哪种情况？
		信息不足，无法确定唯一等级	
		Grade $\in \{1, 2\}$. No exact-grade score target.	

Figure 6: Rule-based synthetic data. (a) Accommodation facts and selection conditions yield *choice*, *noul*, and *score* targets. (b) Missing information permits an insufficient-information *choice* target and a *noul* judgment about whether the grade is determined, but no exact-grade *score* target. Gray boxes contain the generated inputs, shortened for display; each displayed answer has target probability one. The grading rule is fictional.

A.4 Domain Data Sources

The same pipeline in Section 3.1 produces 3,000,000 medical training decisions and 24,000 decisions each for law and finance. Table 7 reports their decision-type counts, and Table 8 lists the legal and financial source contributions. The legal and financial corpora each contain 8,000 decisions per type. Medical data have a larger *noul* share because of their matching and multiple-label tasks.

Table 7: Domain-specific training corpora, counted in decisions by type.

Domain	choice	noul	score	Total
Medical	1,194,496	1,792,223	13,281	3,000,000
Legal	8,000	8,000	8,000	24,000
Finance	8,000	8,000	8,000	24,000

Medical data. The medical corpus draws on 30 source families covering examinations, biomedical understanding, clinical text, and query matching. CMB (Wang et al., 2024) and CMExam (Liu et al., 2023c) supply examination questions, while CBLUE (Zhang et al., 2022) contributes biomedical language-understanding tasks. Multiple-answer examinations become option-membership judgments using Equation 1.

Selection checks the complete tokenized input, using limits of 2,048 tokens overall and 256 tokens for the decision header. A multiple-answer question is retained only when all of its option-wise decisions pass. Tasks with candidate sets that exceed the header budget are excluded. The resulting training set contains 3,000,000 decisions. Its 13,281 *score* examples all use the QTR query–title relevance scale.

The original medical train, validation, and test partitions remain separate. We select 100,000 validation and 20,000 calibration decisions from disjoint groups in the original validation partition. The Medical component of Chinese-Jev Benchmark contains 200,000 decisions from the original test partition.

Table 8: Legal and financial training decisions by source and decision type, after conversion and selection. Each corpus contains 8,000 decisions of each type, totaling 24,000.

Source	choice	noul	score	Total
<i>Legal</i>				
AppealCase (Huang et al., 2025)	1,008	4,081	3,373	8,462
ClaimGen-CN (Zhou et al., 2025)	6,133	2,267	0	8,400
LeCaRDv2 (Li et al., 2024a)	200	200	2,600	3,000
Explicit legal rules	200	400	1,600	2,200
LCR-CN (Zhao et al., 2026)	302	898	0	1,200
eCAIL (Yu et al., 2022)	0	0	400	400
CAPP-130 (Zhu et al., 2023)	157	154	27	338
<i>Financial</i>				
CFLUE (Zhu et al., 2024)	4,100	3,900	0	8,000
FinTruthQA (Xu et al., 2024)	0	1,000	5,000	6,000
Explicit financial rules	400	1,600	3,000	5,000
FinRE (Li et al., 2019)	2,000	1,000	0	3,000
RoleCS (Zhu et al., 2025)	1,500	500	0	2,000

Legal data. ClaimGen-CN provides causes of action for classification (Zhou et al., 2025), and AppealCase provides claim-support labels and changes across appeals (Huang et al., 2025). LeCaRDv2 supplies graded case relevance (Li et al., 2024a), while eCAIL supplies grades based on legal-element overlap (Yu et al., 2022). LCR-CN contributes judgments about conflicts between legal provisions (Zhao et al., 2026), and CAPP-130 adds privacy-policy annotations (Zhu et al., 2023). The resulting tasks span classification, proposition judgments, and ordered scoring through choice, noul, and score.

Financial data. CFLUE supplies professional-examination questions (Zhu et al., 2024), and FinRE supplies entity-relation labels (Li et al., 2019). RoleCS contributes customer-service strategy annotations from upstream model-generated dialogues (Zhu et al., 2025). FinTruthQA provides supervision for financial relevance, answer relevance, and readability (Xu et al., 2024). These annotations support choice and noul judgments, together with score targets that retain the original quality scales.

Legal and financial supplements add decisions governed by explicit fictional rules, including interest, fees, and repayment calculations in finance. Source licenses are retained, including noncommercial terms for ClaimGen-CN, AppealCase, and CFLUE.

A.5 RLCD Objective

The adopted RLCD objective (NandhaKishorM, 2026) evaluates perturbed candidate distributions against the same targets used by cross-entropy.

RLCD samples $G = 4$ perturbations of each decision’s logits. Writing one decision without the batch index, we draw $\eta_g \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$. We center each draw across candidates:

$$\epsilon_g = \eta_g - \frac{\mathbf{1}^\top \eta_g}{K} \mathbf{1}. \quad (8)$$

Adding the centered noise to detached model logits defines the sampled logits:

$$\mathbf{z}_g = \text{sg}(\mathbf{s}) + \epsilon_g, \quad (9)$$

where sg denotes stop-gradient. We convert each sample to a candidate distribution:

$$\mathbf{q}_g = \text{softmax}(\mathbf{z}_g). \quad (10)$$

All perturbations reuse the same model forward pass.

The reward combines log and spherical scores to measure agreement with the target. For ordered score decisions, it also accounts for distance between cumulative distributions through the ranked

probability score:

$$\text{RPS}(\mathbf{q}, \mathbf{y}) = \frac{1}{K-1} \sum_{j=1}^{K-1} \left(\sum_{i=1}^j (q_i - y_i) \right)^2. \quad (11)$$

This term penalizes errors according to the ordering of levels. Combining the three terms gives the reward for each perturbed distribution:

$$R(\mathbf{q}, \mathbf{y}, t) = \sum_{i=1}^K y_i \ell(q_i) + 0.75 \frac{\mathbf{y}^\top \mathbf{q}}{\|\mathbf{q}\|_2} - \mathbb{I}[t = \text{score}] \text{RPS}(\mathbf{q}, \mathbf{y}), \quad (12)$$

where the log score uses the floor $\ell(q_i) = \max(\log q_i, -9.21)$. The RPS term applies only to `score` decisions.

Rewards are centered within each decision’s perturbation group and the resulting advantages are standardized across the batch. Denoting these detached advantages by A_{bg} , the policy-gradient loss is

$$\mathcal{L}_{\text{RL}} = -\frac{1}{BG} \sum_{b=1}^B \sum_{g=1}^G A_{bg} \left[-\frac{\|\mathbf{z}_{bg} - \mathbf{s}_b\|_2^2}{2\sigma^2} \right]. \quad (13)$$

The bracketed term is the Gaussian perturbation surrogate used for the policy-gradient update. With sampled logits and advantages held fixed during differentiation, the update favors perturbations with above-average reward.